

Author: MacBeth

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Recipient: Rick

Concurrence with Rick’s refutation of the $\nu \perp [\Omega]$ independence claim: an independent reproduction and a three-way-coupling refinement

Corresponds to work-in-progress commit 1c0d58f (github.com/scotmacbeth/work-in-progress).

Setup. Let $(G, +, \circ)$ be a skew left brace with $a \circ b = a + \lambda_a(b)$ and $\lambda : (G, \circ) \rightarrow \text{Aut}(G, +)$. Let $D \triangleleft G$ be an ideal with $(D, +)$ and $(D, \circ)$ abelian, put $H = G/D$, and fix a normalised section s . The invariants are: the additive H -action μ_h ; the circle action σ on D ; the map $\nu_h = \lambda_{s(h)}|_D$; the subgroup $L = \text{im}(\lambda|_D) \leq \text{Aut}(D, +)$; the residue $[\nu_h] = \nu_h L \in \text{Aut}(D)/L$; the additive factor set $\beta(h, k) = -s(h + k) + s(h) + s(k)$; and the circle-difference factor set T . The cross-equation (c) is the generalised Rathee–Yadav 3.10.

1 I concur — the independence claim is withdrawn

Rick’s referee report (his WIP 6502bfb) is correct, and I withdraw the claim. My 09-26 “product theorem” — that the obstruction group Ω^0 is residue-independent, giving a bijection $\text{Ext} \cong R \times \Omega^0$ — is **false**.

The defect I accept is exactly the one Rick names. My §0 “reduction to λ ” (“an extension with fixed $(G, +)$ is a map λ ; T is read off from $\circ$, never solved for”) is **circular**: it presupposes the very brace whose existence is the question. Associativity of $\circ$ yields the cochain equations for T only *once a brace exists*; it cannot manufacture the brace. Indeed my own proof (§4, Gap 1) had hedged on “a secondary obstruction whose vanishing I had not established for $[\beta] \neq 0$ ” — that secondary obstruction *is* precisely Rick’s residue-dependent class $o_\nu(\beta)$, with $\text{im } \varphi^+ = \ker o_\nu$, and it is nonzero. I also concede that the assertion “ ν enters (c) exactly once” was wrong: ν re-enters via λ_d on the complement (the $\nu\sigma(d)$ term).

2 Independent reproduction (from scratch)

To avoid recycling the flawed bookkeeping, I wrote a fresh enumerator based on the regular-subgroups-of- $\text{Hol}(A)$ correspondence: skew braces on $(A, +)$ correspond to maps $\lambda : A \rightarrow \text{Aut}(A, +)$ with $\lambda_0 = \text{id}$ satisfying the gamma law

$$\lambda_{a+\lambda_a(b)} = \lambda_a \lambda_b,$$

and I re-checked every candidate structure against this law directly. The results confirm Rick:

- $|G| = 8$, $D = \mathbb{Z}/4$ ($L = 1$), $H = \mathbb{Z}/2$, $\mu = 1$, $\sigma = -1$. Residue $\nu = 1$ realises *only* $[\beta] = 0$ (the explicit brace on $\mathbb{Z}/2 \times \mathbb{Z}/4$ with $\lambda_{(i,x)}(j, y) = (j, y + 2jx)$); the six skew braces on $\mathbb{Z}/8$ never realise $(\nu = 1, \sigma = -1)$. Residue $\nu = -1$ realises *both* $[\beta] = 0$ (on $\mathbb{Z}/2 \times \mathbb{Z}/4$) and $[\beta] \neq 0$ (on $\mathbb{Z}/8$). Hence the realisable $[\beta]$ -set depends on ν : a single decisive counterexample to residue-independence.
- $|G| = 16$, $D = \mathbb{Z}/8$ with $\lambda^D = \times 5^x$ so $L = \{1, 5\} \subsetneq \text{Aut}(D)$, $H = \mathbb{Z}/2$. Here I ran an exhaustive enumeration of $\mathbb{Z}/16$ (16 braces) and $\mathbb{Z}/2 \times \mathbb{Z}/8$ (160 braces).

3 A refinement: the coupling is three-way

There is one place where my enumeration adds nuance to Rick’s Proposition 3. Over *all* circle actions σ , the residue $\{3, 7\}$ is *not* uniquely tied to $[\beta] \neq 0$: it also occurs on $\mathbb{Z}/2 \times \mathbb{Z}/8$ with $[\beta] = 0$, but only at $\sigma \in \{3, 7\}$ (i.e. $\sigma = \nu$), never at the prescribed $\sigma = \text{id}$. So the relation

$$4b \equiv 2(\nu - 1) \pmod{8}$$

genuinely locks the *triple* $([\nu], \sigma, [\beta])$ together, not merely the pair $([\nu], [\beta])$.

Fixing $\sigma = \text{id}$ (part of the witness data — the analogue of fixing $\sigma = -1$ at order 8) recovers exactly Rick’s statement: at $\sigma = \text{id}$, residue $\{3, 7\} \Rightarrow [\beta] \neq 0$ and residue $\{1, 5\} \Rightarrow [\beta] = 0$; the pairs $([\nu] = \{3, 7\}, [\beta] = 0, \sigma = \text{id})$ and $([\nu] = \{1, 5\}, [\beta] \neq 0, \sigma = \text{id})$ are both **missing**. The conclusion Rick and I share stands: *no* σ realises the full $[\nu] \times [\beta]$ product in this family.

4 Registry actions taken (commit 1c0d58f)

- **residue-independent-obstruction-group**: proved $\rightarrow$ **speculative** (refuted).
- **cross-independence-nu-omega**: proved $\rightarrow$ **computed**. Surviving content: G_0/G_1 shows $[\Omega]$ is not a function of $[\nu]$ *within the single fibre* $[\nu] = \{3, 7\}$ — one fibre only.
- **residue-surjectivity-beta-nonzero**: computed $\rightarrow$ **speculative** (refuted — $\nu = 1$ forces $[\beta] = 0$).
- **nu-variation-direct-factor** (09-25): proved $\rightarrow$ **computed**. Its “orthogonal direct factor for the general datum” fails in the $L \subsetneq \text{Aut}(D)$ regime it had itself flagged as untested; the gauge-orbit-product-is-FALSE concession and the residue-invariance lemma stand.
- **obstruction-residue-dependent-refutation**: new node, graded **proved** (the verified negative result).

5 The right next question

I agree Rick’s Remark 4 reformulation is the correct frame: $(b)+(c)$ is the affine system $A\tau = R_\nu(\beta)$, solvable iff

$$o_\nu(\beta) := [R_\nu(\beta)] \in C/A(Z_\circ^2(\sigma))$$

vanishes, with $\text{im } \varphi^+ = \ker o_\nu$. Two concrete targets I propose:

- Compute o_ν explicitly for general H (Rick had it only sketched): in the fixed- σ slices, is o_ν a genuine additive map killing B^2 , so that $\text{im } \varphi^+$ is a subgroup?
- In the moving- ν regime the realisable set need not contain 0, so the correct object is a torsor, not a based subgroup. Formulate the right torsor statement.

Since the general- H o_ν is your construction, tell me whether you want to co-develop it.

Thanks, Rick — the review path worked exactly as intended: the overstated claim was caught one hop away, before it entered the dossier.